



End Semester Examination Autumn 2024-25

Class: **MTech**

Time: 3Hrs

Course Title: Computer Vision

F.M.: 50

Course ID: EC-6603

**This question paper
contains 3 pages.**

Answer all questions.

Figures in the right-hand margin indicate marks.

Describe the following with their applications in computer vision.

1.
 - (a) Difference of Gaussians
 - (b) Thresholding
 - (c) Signatures
 - (d) Distance functions
 - (e) Template matching

2×5

2.
 - (a) Consider a binary image of size 200×200 pixels, with a series of vertical bands of black and white colours as follows: a vertical black band extending from columns 1 to 49, a vertical white band extending from columns 50 to 100, a vertical black band extending from columns 101 to 149 and a vertical white band extending from columns 150 to 200.
 - i. Obtain the co-occurrence matrix of this image using the position operator “one pixel to the right.”
 - ii. Normalize this matrix so that its elements become probability estimates.
 - iii. Use your matrix from (ii) to compute the descriptors: Contrast, Uniformity, Homogeneity, and Entropy.
 - (b) Describe Fourier descriptor and its applications in computer vision.

6
2

3.
 - (a) Discuss Otsu’s method of segmentation. What is the concept behind it? What is its significance? Clearly explain the algorithm. When does it fail?
 - (b) Determine the chain code and shape number of all possible shapes of order 10.
 - (c) Image subtraction is often used in industrial applications for detecting missing components in product assembly. The approach is to store a “golden” image that corresponds to a correct assembly; this image is then subtracted from incoming images of the same product. Ideally, the differences would be zero if the new products are assembled correctly. Difference images for products with missing components would be nonzero in the area where they differ from the golden image. What conditions do you think have to be met in practice for this method to work?

4
2
2

4. (a) In a medical test to identify sick persons, when 120 persons were tested, it was found that only 22 out of 26 truly sick persons were detected as unhealthy, while 86 out of 94 healthy people were identified as healthy. Determine its confusion matrix. Find the accuracy, precision, sensitivity, specificity, and F1 score of the testing system. 4
- (b) Find the Euler number for images of objects looking like: “A”, “B” “C”, and “D” and draw your conclusions. 2
- (c) Sketch the gradient and Laplacian of each profile shown in Fig.1. 2

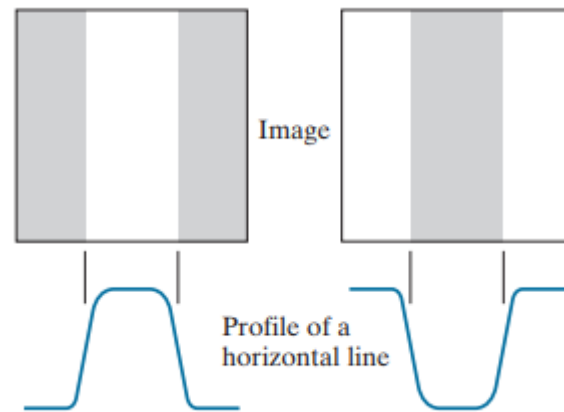


Fig.1

5. (a) The speed of a bullet in flight is to be estimated by using high-speed imaging techniques. The method of choice involves the use of a CCD camera and flash that exposes the scene for K seconds. The bullet is 2.5 cm long, 1 cm wide, and its range of speed is 500 ± 100 m/s. The camera optics produces an image in which the bullet occupies 10% of the horizontal resolution of a 512×512 digital image.
 - i. Determine the maximum value of K that will guarantee that the blur from motion does not exceed 2 pixels.
 - ii. Determine the minimum number of frames per second that would have to be acquired in order to guarantee that at least two complete images of the bullet are obtained during its path through the field of view of the camera.
 - iii. Propose a segmentation procedure for automatically extracting the bullet from a sequence of frames.
 - iv. Propose a method for automatically determining the speed of the bullet. 6
- (b) A company that bottles a variety of industrial chemicals employs you to design an approach for detecting when bottles of their product are not full. As they move along a conveyor line past an automatic filling and capping station, the bottles appear as shown in Fig. 2. A bottle is considered imperfectly filled when the level of the liquid is below the midway point between the bottom of the neck and the shoulder of the bottle. The shoulder is defined as the intersection of the sides and slanted portions of the bottle. The bottles move at a high rate of speed, but the company has an imaging system equipped with an illumination flash front end that effectively stops motion, so you will be given images that look very close to the sample shown here. Propose a solution for detecting bottles that are not filled properly. State clearly all assumptions that you make and that are likely to

impact the solution you propose.

2



Fig. 2

- 6 (a) Use the k-means algorithm employing Euclidean distance to cluster the following 8 samples into 3 clusters: $A1 = (2,10)$, $A2 = (2,5)$, $A3 = (8,4)$, $A4 = (5,8)$, $A5 = (7,5)$, $A6 = (6,4)$, $A7 = (1,2)$, $A8 = (4,9)$. Suppose that the initial seeds (centers of each cluster) are $A1$, $A4$ and $A7$. Run the k-means algorithm for 1 iteration only. At the end of this iteration show:
- The new clusters (i.e. the samples belonging to each cluster)
 - The centers of the new clusters
 - Draw a 10 by 10 space with all the 8 points and show the clusters after the first iteration and the new centroids.
 - How many more iterations are needed to converge? Draw the result for the final iteration.
- (b) An automated pharmaceutical plant uses image processing to measure the shapes of medication tablets for the purpose of quality control. The segmentation stage of the system is based on Otsu's method. The speed of the inspection lines is so high that a very high rate flash illumination is required to "stop" motion. When new, the illumination lamps project a uniform pattern of light. However, as the lamps age, the illumination pattern deteriorates as a function of time and spatial coordinates according to the equation:

6

$$i(x,y) = A(t) - t^2 e^{-[(x-\frac{M}{2})^2 + (y-\frac{N}{2})^2]}$$

where $(M/2, N/2)$ is the center of the viewing area and t is time measured in increments of months. The lamps are still experimental and the behavior of $A(t)$ is not fully understood by the manufacturer. All that is known is that, during the life of the lamps, $A(t)$ is always greater than the negative component in the preceding equation because illumination cannot be negative. It has been observed that Otsu's algorithm works well when the lamps are new, and their pattern of illumination is nearly constant over the entire image. However, segmentation performance deteriorates with time. Being experimental, the lamps are exceptionally expensive, so you are employed as a consultant to help solve the problem to compensate for the changes in illumination, and thus extend the useful life of the lamps. You are given flexibility to install any special markers or other visual cues in the viewing area of the imaging cameras. Propose a solution in sufficient detail.

2